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Target layer state estimation in multilayer complex dynamical networks using functional observability

Yayong Wu^{a,b}, Xinwei Wang^{a,b}, Guo-Ping Jiang^{a,b,*}, Mengqi Gu^{a,b}

^a College of Automation and College of Artificial Intelligence, Nanjing University of Posts and Telecommunications, Nanjing 210023, PR China

^b Jiangsu Engineering Lab for IOT Intelligent Robots (IOTRobot), Nanjing 210023, PR China

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Abstract

In large-scale complex dynamical networks, it is significant to estimate the states of target nodes with only a part of measured nodes. Meanwhile, multilayer complex dynamical networks exist widely in society and engineering. Therefore, it has important theoretic meaning and practical value to study the state estimation of target nodes in multilayer complex dynamical networks with limited node measurements. In this paper, with the measurable state information of a portion of nodes in one layer in the multilayer complex dynamical network, the state estimation of target nodes in other layers is studied. First, we build the model of the multilayer complex dynamical network which includes some target nodes and sensor nodes. Second, auxiliary nodes are selected by using the maximum matching principle in graph theory to construct the augmented node set. Third, we discuss the relationship between the minimum number of auxiliary nodes and interlayer connection probability in the multilayer complex dynamical network. Forth, an appropriate functional state observer is designed with limited number of measured nodes according to a typical model-based algorithm. Finally, numerical simulations are given to demonstrate the accuracy of the proposed method. The proposed method can achieve the accurate estimation with less placement of observers and fewer computational costs in the multilayer complex dynamical network.

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* Corresponding author at: College of Automation and College of Artificial Intelligence, Nanjing University of Posts and Telecommunications, Nanjing 210023, PR China.

E-mail addresses: 2019050110@njupt.edu.cn (Y. Wu), xwwang@njupt.edu.cn (X. Wang), jianggp@njupt.edu.cn (G.-P. Jiang), mengqigu@njupt.edu.cn (M. Gu).

1. Introduction

Complex dynamical networks have received great attention over the past decades because of its accurate depiction in different fields such as transportation networks, power grids, epidemic spreading and so on [1–3]. In social and engineering networks, in order to monitor the behavior of the network and judge the network faults timely, it is often necessary to obtain the state information of nodes in the network [4–7]. Therefore, how to use the known state information of a part of nodes to estimate that of other unknown nodes has become a hotspot in the research of complex dynamical networks.

Nowadays, most publications regarding the state estimation are focused on single-layer networks [8–16]. Nguyen et al. studied the distributed state estimation problem of the network and presented the design method of distributed state estimator in Nguyen and Kling [8]. Targui et al. estimated the state of dynamical nodes in communication networks by using the state observer in Targui et al. [9]. Fan et al. proposed topological identification criteria for complex dynamical networks with chaotic nodes by using adaptive observer method in Fan et al. [10]. Researchers also took packet loss, time delay and other external disturbances into consideration in the state estimation. Wu et al. studied the state estimation problem of the discrete time complex dynamical network with packet loss between the original network and the state observer network in Wu et al. [11]. Wang et al. studied the state estimation problem of a continuous-time linear time-invariant system with unknown external input disturbance and measurement noise in Wang et al. [12]. Wang et al. proposed a new identification controller to detect the topology of weighted complex dynamical networks considering incomplete measurements in Wang et al. [13]. Wu et al. used the maximum matching principle of graph theory to study the observability problem of the complex dynamical network with fully matched root strongly connected components in Wu et al. [14]. Wan et al. studied the issue of finite-time H_∞ state estimation for a class of discrete-time nonlinear two-time-scale complex networks under the stochastic communication protocol in Wan et al. [15]. Wang et al. investigated the consensus problem of a continuous-time uncertain multi-agent system (MAS) with time-varying interval uncertainties in the system matrix and unknown initial states in Wang et al. [16]. The above researches only paid attention to the state estimation of the single-layer network with all the node being measured.

As the complex dynamical network grows larger, it is prohibitive to set up sensors for each node to measure the state information considering either economic costs or physical constraints [17,18]. For example, it is difficult to measure the states of the tens of thousands of neurons in the human brain; In social networks, our ability to acquire the information of each individual is physically limited due to the large number of individuals. At the same time, in many practical problems, it is not necessary to estimate the state information of all nodes in the large-scale network, we usually only pay attention to the state information of some interested target nodes [19]. For example, in the decentralized control strategy in the network system, each controller only needs feedback signals from a small number of nodes in the corresponding control domain [20]. Therefore, it is necessary to study the state estimation of the interested nodes based on the limited node measurement in the network.

In recent years, the state estimation of complex dynamical networks based on partial nodes has been concerned by many researchers [21–24]. Liu et al. designed a state estimator based on partial node measurements for a class of complex dynamical networks with unbounded distributed delay and energy-bounded measurement noises in Liu et al. [21]. Li et al. stud-

ied the problem of ensuring state estimation under the condition that some neurons can be measured for artificial neural networks with randomly time delays in Li et al. [22]. Liu et al. determined the minimum sensor node set to ensure observability of complex networks based on the controllability graph theory method and constructed state observer to realize state estimation of complex networks in Liu et al. [23]. Liu's achievement is an important connection between graph theory and control of complex networks. However, even if a minimum set of sensor nodes is found, the state observer will have the same dimension as the entire network. In real networks, it is difficult to set up sensors for each node to measure the state information considering either economic costs or physical constraints. Montanari et al. developed a graph-theoretic characterization of functional observability and associated algorithms for sensor placement and observer design in Montanari et al. [24]. Compared with Liu's results, it can estimate the state information of target nodes of a large-scale dynamical network accurately with less placement of sensors and observers.

With the research of complex dynamical networks, researchers gradually realize that most networks in real society and engineering do not exist independently, but are related with other networks in the structure or function [25,26]. Single-layer network structures can not describe the dynamical characteristics of many real systems or networks accurately. For example, in smart grid networks, the power layer and the information layer are highly integrated to realize reliable operation in the grid [25]; in epidemic spreading networks, as diseases spread through the networks of contacts, the information about diseases also spreads across social networks, which will influence the degree of the disease spreading [26]. Multilayer complex dynamical networks enable us to characterize more real networks and have become a significant topic of research in complex dynamical networks.

At present, the studies of multilayer complex dynamical networks have received great attention, mainly focusing on the synchronization [27,28], diffusion dynamics [29], topology identification [30] and dynamic cascade failure [31]. However, there is little research on the state estimation of multilayer complex dynamical networks. The condition of observability is the prerequisite of state estimation for the linear system, and there is a dual relationship between observability and controllability. Several publications have studied the controllability of multilayer complex dynamical networks. Chapman et al. extended the controllability of factor networks to composite networks and obtained the necessary and sufficient conditions for system controllability in multilayer networks in Chapman et al. [32]. Hao et al. studied the controllability of a special class of homogeneous multilayer networks in Hao et al. [33]. Chen et al. developed the controllability of hyper-graph network according to the theory of polynomial control and derived the Kalman rank condition to determine the minimum number of control inputs to ensure controllability in Chen et al. [34]. Wu et al. studied the state controllability of multilayer networks with interlayer couplings and obtained some necessary and sufficient conditions for the network controllability in Wu et al. [35]. Jiang et al. provided sufficient and necessary conditions for the controllability of multilayer homogeneous networks in the form of matrix equations, as well as sufficient or necessary conditions for the controllability of networks with special structures in the intralayer network in Jiang et al. [36]. Cao et al. investigated the controllability of multilayer complex networks with high-dimensional node states and discussed the influence of node heterogeneity, interlayer coupling, and intralayer coupling on the controllability of the network in Cao et al. [37]. However, despite the duality between controllability and observability, such duality does not hold between target controllability and target observability [24]. Therefore, it is significant to estimate the

state information of target nodes on other layers by using the known state information of some nodes on one layer in the multilayer complex dynamical network.

In this paper, we study the state estimation of target nodes in multilayer complex dynamical networks with limited node measurements. First, a multilayer complex dynamical network model with arbitrary connections between interlayer is constructed, in which the state information of some nodes in one layer has been measured by sensors. These nodes are called sensor nodes. The state information of target nodes in other layers need to be estimated. Second, based on the topological structure and the known state information of some nodes in one layer, the auxiliary nodes are found by using the maximum matching principle of graph theory to construct an augmented node set. Third, the relationship between the minimum number of auxiliary nodes in the functional observer design and interlayer connection probability in the multilayer complex dynamical network has been analyzed. Forth, an appropriate functional state observer is designed based on a typical algorithm to estimate the state information for target nodes in other layers. Finally, the effectiveness of the proposed method is demonstrated from some numerical simulations. We only place observers on the augmented node set, which can achieve the accurate estimation with less placement of observers and fewer computational costs.

The major contributions of this paper are summarized as follows:

- (1) We study the problem of the target layer state estimation in multilayer complex dynamical networks by using functional observability based on the actual needs of the engineering.
- (2) We analyze the relationship between the minimum number of auxiliary nodes in the functional observer design and interlayer connection probability in the multilayer complex dynamical network.
- (3) The proposed method can achieve the accurate estimation with less placement of observers and fewer computational costs in the multilayer complex dynamical network.

The rest of this article is organized as follows. The model of the multilayer complex dynamical network which includes some target nodes and some sensors nodes is established in Section 2. In Section 3, the rank-based condition of state estimation of target nodes is analyzed. The augmented node set is constructed based on the maximum matching principle in Section 4. The relationship between the minimum number of auxiliary nodes and interlayer connection probability is analyzed in Section 5. Functional state observer is determined in Section 6. Some numerical simulations are developed to demonstrate the accuracy of the proposed method in Section 7. Last, the conclusions are drawn in Section 8.

2. The model of the multilayer complex dynamical network

Consider a multilayer complex dynamical network consisting of L layers and each layer contains $N^{(l)}$ nodes, ($l = (1, 2, \dots, L)$). N is the total number of nodes in the network. The dynamical equation can be described as

$$\begin{cases} \dot{x}_i^l(t) = A_i^l(x_i^l(t)) + \sum_{j=1}^{N^{(l)}} d_{ij}^l H^l x_j^l(t) + \sum_{R=1, R \neq l}^L \sum_{j=1}^{N^{(l)}} d_{ij}^{lR} H^{lR} x_j^R(t) + B_i^l u_i^l(t), \\ y_i^l(t) = C_i^l x_i^l(t), \end{cases} \quad (1)$$

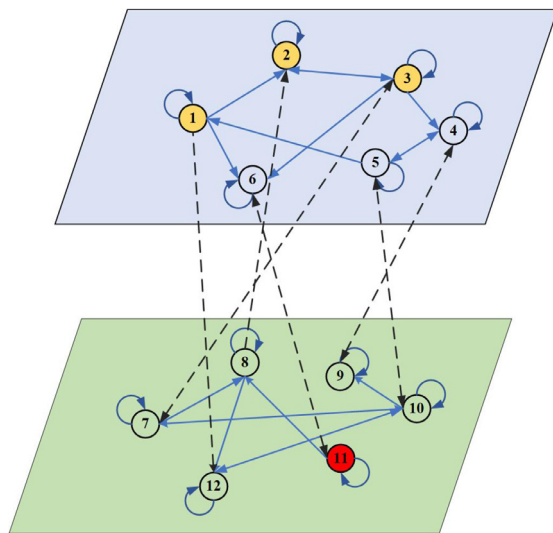


Fig. 1. The model of the multilayer complex dynamical network to study the target layer state estimation using functional observability.

where $(i = (1, 2, \dots, N^{(l)}))$, $x_i^l \in R^n$ and $u_i^l \in R^m$ are the n -dimensional state vector and the m -dimensional control input of the i th node on the layer l respectively. $y_i^l \in R^q$ represents the output of the i th node on the layer l , that is the direct measurement from the available sensors in the network. $A_i^l \in R^{n \times n}$, $B_i^l \in R^{n \times m}$ and $C_i^l \in R^{q \times n}$ are the state matrix, input matrix and output matrix of the i th node on the layer l respectively. $D^l = [d_{ij}^l]_{N^{(l)} \times N^{(l)}}$ is the intralayer configuration matrix of the network on the layer l , if there is a link from node j to node i , $d_{ij}^l > 0$, otherwise, $d_{ij}^l = 0$. $D^{lR} = [d_{ij}^{lR}]_{N^{(l)} \times N^{(R)}}$ is the interlayer configuration matrix of the network, if there is a link from node j on the layer R to node i on the layer l , $d_{ij}^{lR} > 0$, otherwise, $d_{ij}^{lR} = 0$. H^l and H^{lR} account for the inner coupling matrix of the intralayer network and the interlayer network.

The two-layer network, as a special case of the multilayer complex dynamical network, possesses different correlations between nodes in different layers, which is a suitable model to study the multilayer complex dynamical network. In this paper, we take a two-layer complex dynamical network and assume every node with one-dimensional which is shown in Fig. 1 as an example. Other types of multilayer networks can also achieve the state estimation of target nodes according to the proposed method. The nodes which marked in yellow on one layer are the sensor nodes and their information can be measured directly from sensors. The node marked in red on another layer is the target node. In the one-dimensional network, the output information of one node is the same as its state information. Our purpose is to achieve the state estimation of the red node with yellow measured nodes in one layer.

Combined with Fig. 1, the Eq. (1) can be transformed into Eq. (2)

$$\begin{cases} \dot{X}(t) = \varphi X(t) + \psi U(t), \\ Y(t) = \xi X(t), \end{cases} \quad (2)$$

where $X = (x_1, x_2, \dots, x_N)^T$, $U = (u_1, u_2, \dots, u_N)^T$, $Y = (y_1, y_2, \dots, y_q)^T$, $\varphi \in R^{N \times N}$, $\psi \in R^{N \times m}$, $\xi \in R^{q \times N}$, q is the number of sensor nodes.

$$\varphi = \begin{bmatrix} * & 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & * & * & 0 & 0 & 0 & 0 & * & 0 & 0 & 0 & 0 \\ 0 & * & * & 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & * & * & * & 0 & 0 & 0 & * & 0 & 0 & 0 \\ 0 & 0 & 0 & * & * & 0 & 0 & 0 & 0 & * & 0 & 0 \\ & 0 & * & 0 & 0 & * & 0 & 0 & 0 & 0 & * & 0 \\ 0 & 0 & * & 0 & 0 & 0 & * & 0 & 0 & * & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & * & * & 0 & 0 & * & 0 \\ 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & * & * & 0 & 0 \\ 0 & 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & * & 0 \\ & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & * & 0 & * \end{bmatrix}. \quad (3)$$

In this model, only node 1, 2 and 3 in the first layer are placed with sensors. Therefore, we can only obtain the output information of node 1, 2 and 3 in the first layer. Then one has

$$\xi = \left[\begin{array}{cccccc|cccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

In large-scale networks, it is not necessary for us to estimate the state information of all nodes. Only a set of target nodes will be paid attention to. Given a matrix $F \in R^{r \times N}$, the target nodes $z(t) = FX(t) \in R^r$, where r can be much smaller than N . In Fig. 1, only node 11 in the second layer is of interest, then we set

$$F = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0].$$

Our purpose is to estimate the state information of target nodes $z(t)$ from the limited number of measured nodes.

Remark 1. Each row of sensor node matrix ξ or target node matrix F has only one nonzero entry. For example, if we want to obtain the state information of node 11 and node 12. we set

$$F = \left[\begin{array}{cccccc|cccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right].$$

If we set $F = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1]$, what we get is the sum of the state information of node 11 and node 12.

Remark 2. If the target nodes are distributed in each layer of the network. The proposed method is still applicable. In this case, we only need to change the matrix F so that the elements corresponding to the target nodes are set to 1. For example, if the node 4 in the first layer and the node 11 in the second layer are target nodes in Fig. 1. Then the matrix F will be set as follows.

$$F = \left[\begin{array}{cccccc|cccccc} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{array} \right].$$

3. The rank-based condition of state estimation of target nodes

If F is desirable, we can reconstruct the target nodes $z(t)$ from the direct measurement $Y(t)$ [38]. If and only if condition in Eq. (4) is satisfied [39–41].

$$\text{rank} \begin{bmatrix} \xi \\ \xi\varphi \\ \vdots \\ \xi\varphi^n \\ F \end{bmatrix} = \text{rank} \begin{bmatrix} \xi \\ \xi\varphi \\ \vdots \\ \xi\varphi^n \end{bmatrix}. \quad (4)$$

However, the condition in Eq. (4) only guarantees the existence of a functional observer. To design a functional state observer, the two additional conditions in Eqs. (5) and (6) must be held [41].

$$\text{rank} \begin{bmatrix} \xi \\ \xi\varphi \\ F_0 \\ F_0\varphi \end{bmatrix} = \text{rank} \begin{bmatrix} \xi \\ \xi\varphi \\ F_0 \end{bmatrix}. \quad (5)$$

$$\text{rank} \begin{bmatrix} \lambda_i F_0 - F_0\varphi \\ \xi \\ \xi\varphi \end{bmatrix} = \text{rank} \begin{bmatrix} \xi\varphi \\ \xi \\ F_0 \end{bmatrix}. \quad (6)$$

where $F_0 \in R^{r_0 \times N}$ is the augmented node set and λ_i is every eigenvalue of φ .

If a triple (φ, ξ, F) satisfies Eq. (4), which means there exists a matrix F_0 whose row space contains that of F that satisfies Eqs. (5) and (6) for (φ, ξ, F_0) . A functional state observer of order $r_0 \geq r$ can be designed after a suitable matrix F_0 is found. If $r_0 > r$, only placing observers to target nodes is not enough. We still need to find appropriate nodes (called auxiliary nodes) on this layer or other layers to place with observers in the multilayer complex dynamical network. The target nodes and auxiliary nodes construct the augmented node set F_0 to satisfy the condition in Eqs. (5) and (6). Our next task turns to finding auxiliary nodes.

4. Construct the augmented node set

Previous studies on functional observability mainly focus on find the existence conditions for functional observability, that is the rank-based conditions mentioned in Eqs. (4)–(6) in this paper. Few researches have studied how to find auxiliary nodes through arithmetic methods [42,43]. At the same time, the arithmetic methods have a large amount of computation which is not suitable for large-scale or high-dimensional networks. In order to overcome this difficulty, researchers convert the rank-based condition into its equivalent graph theory approach that explicitly leverages the structure of the network, which is a great approach to find the auxiliary nodes for large-scale or high-dimensional networks [23,24]. In this paper, we select the auxiliary nodes by using the maximum matching method in graph theory according to the principle proposed in Montanari et al. [24].

Theorem 1. A system (φ, ξ, F) is structurally functionally observable if and only if the following conditions are satisfied: (1) There exists a direct path from each target node on one layer to some sensor nodes on other layers. (2) No target node is an element of a minimal subset of nodes with a dilation.

Remark 3. The definitions of path and dilation are quoted from Lin's definition in Lin [44].

Remark 4. We cite the algorithm in Montanari et al. [24] to determine a minimum sensor set such that there are direct paths from every target node to some sensor nodes.

Corollary 1. *If every target node has a self-link and Eq. (5) is true for a system (φ, ξ, F) , then Eq. (6) is true for the same system.*

The above theorem and corollaries are strictly proved in Montanari et al. [24] and the Algorithm 1 to find the appropriate auxiliary nodes are as follows.

Algorithm 1.

Step 1: Initialize $F_0 \leftarrow F$, $M_1 \leftarrow \emptyset$, $M_2 \leftarrow \emptyset$, update $G \leftarrow [\xi^T (\xi\varphi)^T F_0^T]$, build the correspond bipartite graph $\Pi(V, V')$ for matrix G . The elements in V corresponding to a row of G and the elements in V' corresponding to a column of G .

Step 2: Find the maximum matching set ε_m with $\Pi(V, V')$ by Hungarian algorithm. If $v'_i \in V'$ and $v'_i \in \varepsilon_m$, then update the set of matched nodes $M_1 \leftarrow M_1 \cup v'_i$.

Step 3: Define the candidate node set of auxiliary nodes $K = M_2 \setminus M_1$. If v'_j is a nonzero element in $[F_0\varphi]_{ij}$, then update $M_2 \leftarrow M_2 \cup v'_j$. $v'_k \in K$, if $v'_k \in M_2$ and $v'_k \notin M_1$.

Step 4: Select an element in v'_k in K and update $F_0 \leftarrow [F_0^T (F')^T]^T$, where $F' \in R^{1 \times N}$ and $[F']_{1j} = 1$ if $j = k$, otherwise, $[F']_{1j} = 0$. Repeat steps 1–3. When $K = \emptyset$, the process terminates.

Corollary 2. *A system (φ, ξ, F) is structurally functionally observable, Algorithm 1 returns a matrix F_0 with the smallest order such that Eq. (5) is satisfied subject to the constraint that F_0 has only one nonzero entry per row.*

Take the two-layer network in Fig. 1 as an example, assume that the nonzero elements in all matrices are set to 1. The process of finding auxiliary nodes in the two-layer network can be seen in Fig. 2. With Step 2, the node 1, 2, 3, 5 in the first layer and node 7, 8, 11 in the second layer are matched, then $M_1 = \{1, 2, 3, 5, 7, 8, 11\}$. M_2 is a nonzero element set in $[F\varphi]_{ij}$ and $M_2 = \{6, 11\}$. The node 6 in the first layer is a nonzero element in $[F\varphi]_{ij}$ but unmatched node in M_1 . Then the candidate node set of auxiliary nodes $K = \{6\}$ is obtained. We take the node 6 in the first layer as an auxiliary node. Repeat Steps 1–4 until $K = \emptyset$.

According to the above steps, we only need to select the node 6 in the first layer as an auxiliary node to satisfy the condition in Eqs. (5) and (6). It is not necessary to place observers for all nodes with unknown states.

5. The relationship between the minimum number of auxiliary nodes and interlayer connection probability

From the above discussion, the augmented node set F_0 contains target nodes and auxiliary nodes. We finally estimate the state information of target nodes as well as auxiliary nodes. However, our goal is just to estimate the state of target nodes but not the auxiliary nodes. We wish to minimize the number of auxiliary nodes to reduce the computational costs. The interlayer connection is the main channel of information interaction in the multilayer complex dynamical network. In this section, we will discuss the relationship between the minimum number of auxiliary nodes and interlayer connection probability in a general two-layer complex dynamical network.

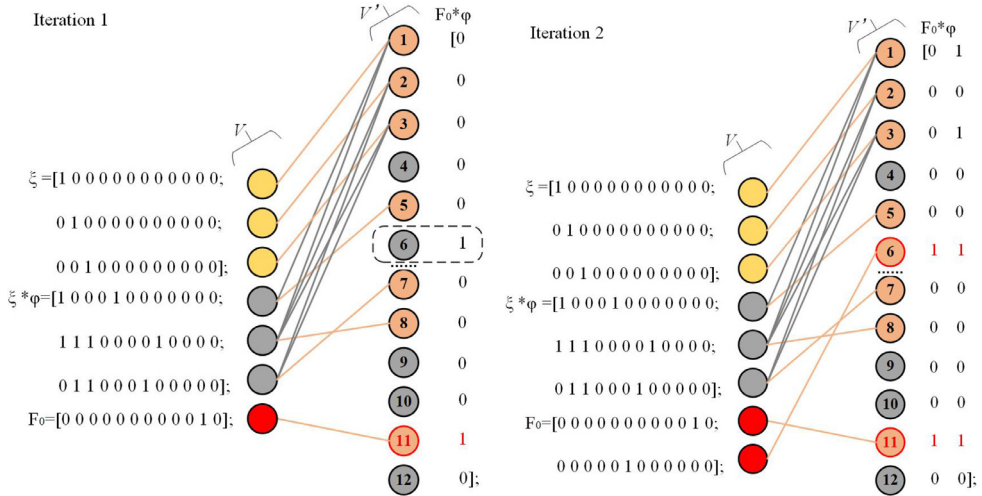


Fig. 2. The process of finding auxiliary nodes in the two-layer complex dynamical network. The matched nodes are highlighted in orange, the selected auxiliary nodes in each iteration are framed in dotted boxes and the nodes which highlighted in orange and red circle are auxiliary nodes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Assume the two-layer network contains 500 nodes, the intralayer networks are randomly generated by scale-free (SF) and small-world (SW) algorithms respectively. The SF network is generated with the Barabási–Albert model, the number of new links connected to existing nodes is 6. The SW network is generated with the Newman–Watts model, the probability of adding a new edge between two existing nodes is 0.5. The directed interlayer network is generated by random network with the Erdos–Rényi model, where p is the probability of connection between nodes.

In the results, N is the number of total nodes in the network, $N^{(1)}$ is the number of nodes in the first layer, $N^{(2)}$ is the number of nodes in the second layer, q is the number of sensor nodes, r is the number of target nodes, r_0 is the order of the functional observer, $\Delta r = r_0 - r$ is the number of auxiliary nodes. In each simulation, let the sensor nodes reside on the first layer and the target nodes reside on the second layer. Sensor nodes and target nodes are randomly placed in the corresponding layer. The sensor placement is implemented by the following steps: first find the minimum set of sensor nodes for functional observability, then increase q with randomly placed sensor nodes. When we study the relationship between p and Δr with different proportions of the number of sensor nodes, we set $r = 0.3N^{(2)}$, and with different proportions of the number of target nodes, we set $q = 0.3N^{(1)}$. To make the simulation more representative, 50 independent detection results are averaged.

5.1. SF-SF

We first consider a two-layer network, the first layer and the second layer are both SF networks. Fig. 3(a) shows the number of auxiliary nodes versus the probability of connection between nodes with different proportions of sensor nodes. Overall, a larger sensor set q leads to a lower number of Δr in the different probabilities of connection between interlayer nodes. Taking $q/N^{(1)} = 0.5$ as the reference, when $q/N^{(1)}$ is less than 0.5, Δr increases with the

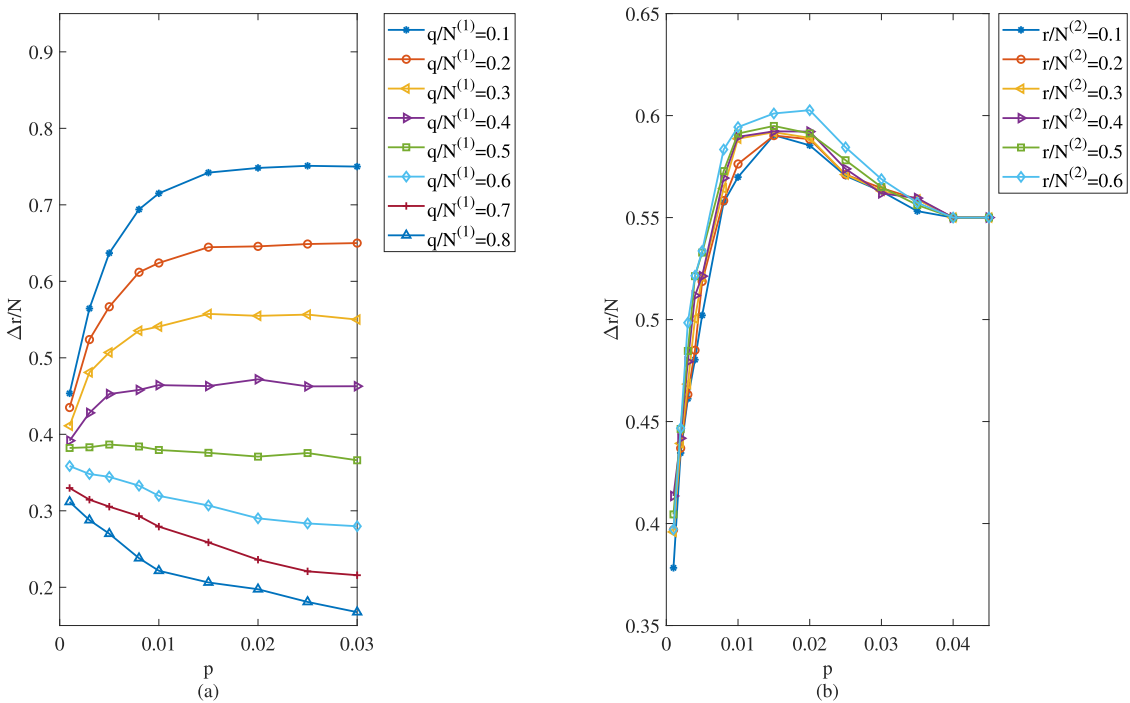


Fig. 3. The minimum number of auxiliary nodes in a two-layer complex dynamical network (SF-SF) (a) The number of auxiliary nodes Δr versus the probability of connection between interlayer nodes with different proportions of sensor nodes. (b) The number of auxiliary nodes Δr versus the probability of connection between interlayer nodes with different proportions of target nodes.

increase of p and finally reaches a stable value. When $q/N^{(1)}$ is more than 0.5, Δr decreases continuously with the increase of p . Fig. 3(b) shows the number of auxiliary nodes versus the probability of connection between nodes with different proportions of target nodes. It can be seen that the number of target nodes r has little influence on the number of auxiliary nodes Δr with the same p . With the increase of p , Δr also increases. When $p \approx 0.015$, Δr reaches peak value. Then Δr decreases gradually and finally reaches a fixed value.

5.2. SW-SW

We then consider a two-layer network whose first layer and second layer are both generated by SW algorithm. Fig. 4(a) and (b) illustrate Δr versus p with different proportions of sensor nodes and target nodes respectively. From simulations, when the number of sensor nodes is small, we need a large number of auxiliary nodes. With the increase of q and p , Δr decreases. When p reaches about 0.02, Δr approaches a stable value in Fig. 4(a). The number of target nodes r also has little influence on Δr . With the increase of p , Δr decreases and finally reaches a fixed value in different proportions of target nodes.

5.3. SF-SW

In this subsection, the two-layer complex dynamical network is considered, of which the first layer is generated by SF network and the second layer is generated by SW network. The simulation results are shown in Fig. 5. Obviously, it has the same tendency with the two-layer network of the first layer and the second layer are both SF networks. In Fig. 5(a), we also take $q/N^{(1)} = 0.5$ as the reference, Δr increases with the increase of p when $q/N^{(1)}$ is less than 0.5. When $q/N^{(1)}$ is more than 0.5, Δr decreases with the increase of p . In Fig. 5(b), when p is about 0.015, Δr reaches maximum value.

5.4. SW-SF

We finally consider a two-layer complex dynamical network, the first layer is generated by SW network and the second layer is generated by SF network. From Fig. 6, it has the same tendency with the two-layer network of the first layer and the second layer are both SW networks. As with the SW-SW structure, sensor nodes are both arranged on the SW layer, Fig. 6 has similar conclusions to Fig. 4, which means the network structure of the layer where the sensor nodes reside plays a dominant role in the influence of the interlayer connection probability on the minimum number of auxiliary nodes.

According to the above four simulations, we obtain the following conclusions about the relationship between the minimum number of auxiliary nodes and interlayer connection probability:

- (1) With the same probability of connection between interlayer nodes, a larger sensor set q leads to a lower number of Δr , and the number of target nodes r has little influence on Δr .
- (2) The network structure of the layer where the sensor nodes reside determines the influence of the interlayer connection probability on the minimum number of auxiliary nodes.

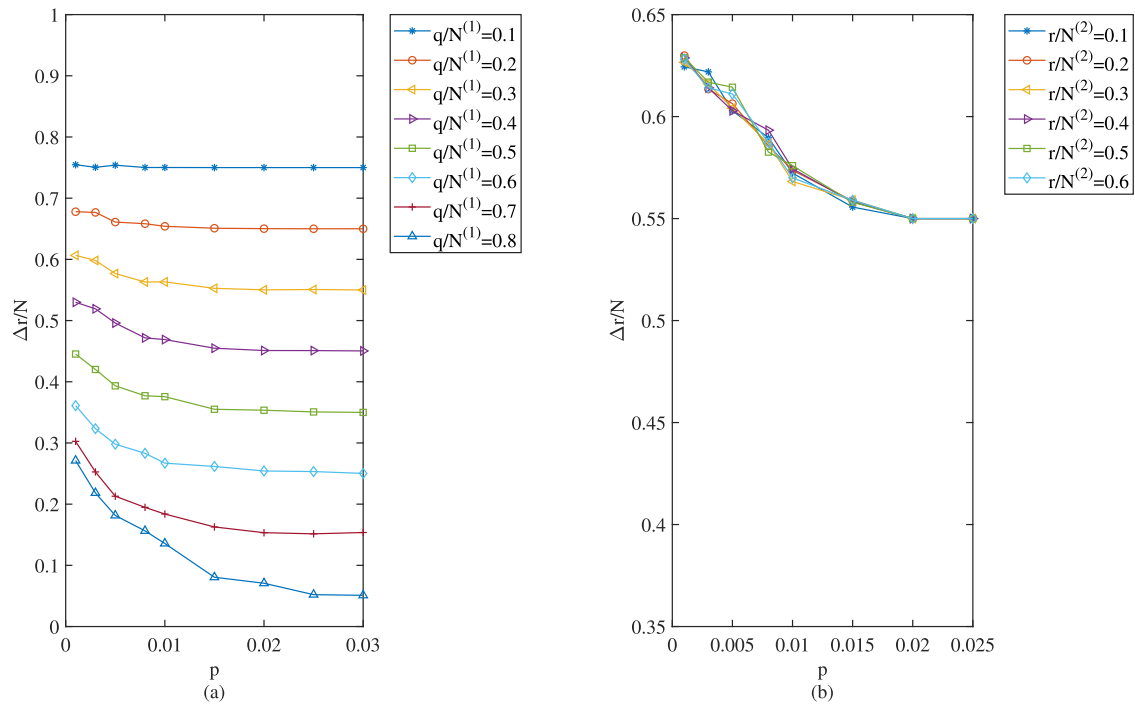


Fig. 4. The minimum number of auxiliary nodes in a two-layer complex dynamical network (SW-SW) (a) The number of auxiliary nodes Δr versus the probability of connection between interlayer nodes with different proportions of sensor nodes. (b) The number of auxiliary nodes Δr versus the probability of connection between interlayer nodes with different proportions of target nodes.

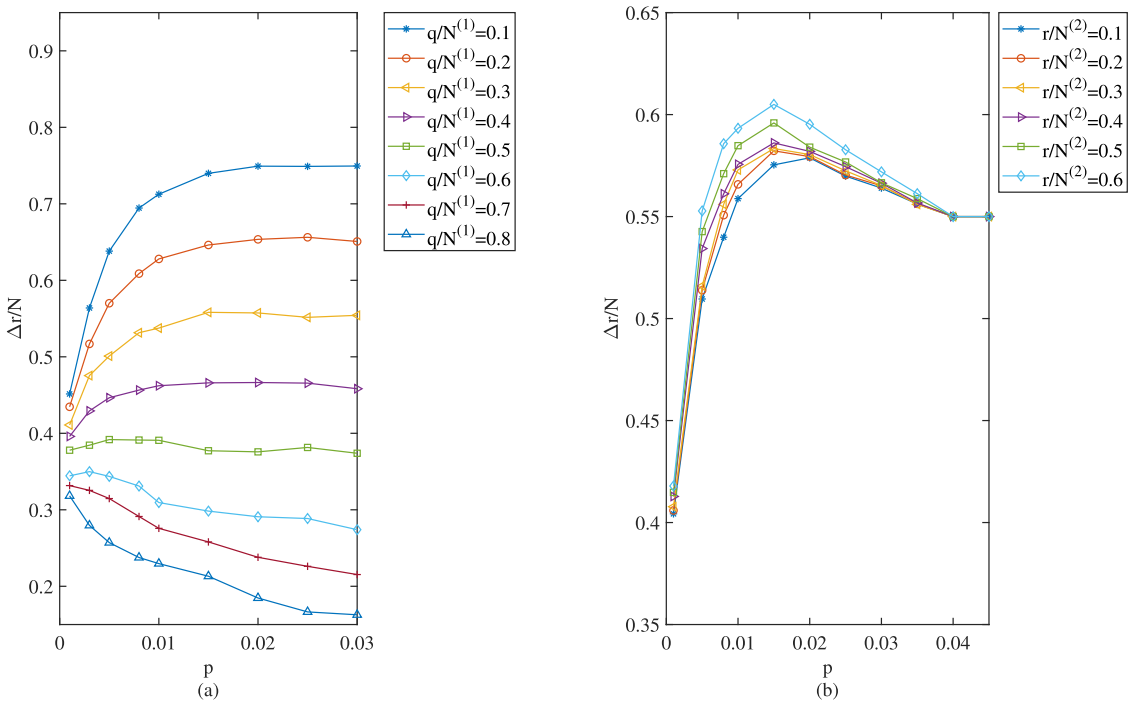


Fig. 5. The minimum number of auxiliary nodes in a two-layer complex dynamical network (SF-SW) (a) The number of auxiliary nodes Δr versus the probability of connection between interlayer nodes with different proportions of sensor nodes. (b) The number of auxiliary nodes Δr versus the probability of connection between interlayer nodes with different proportions of target nodes.

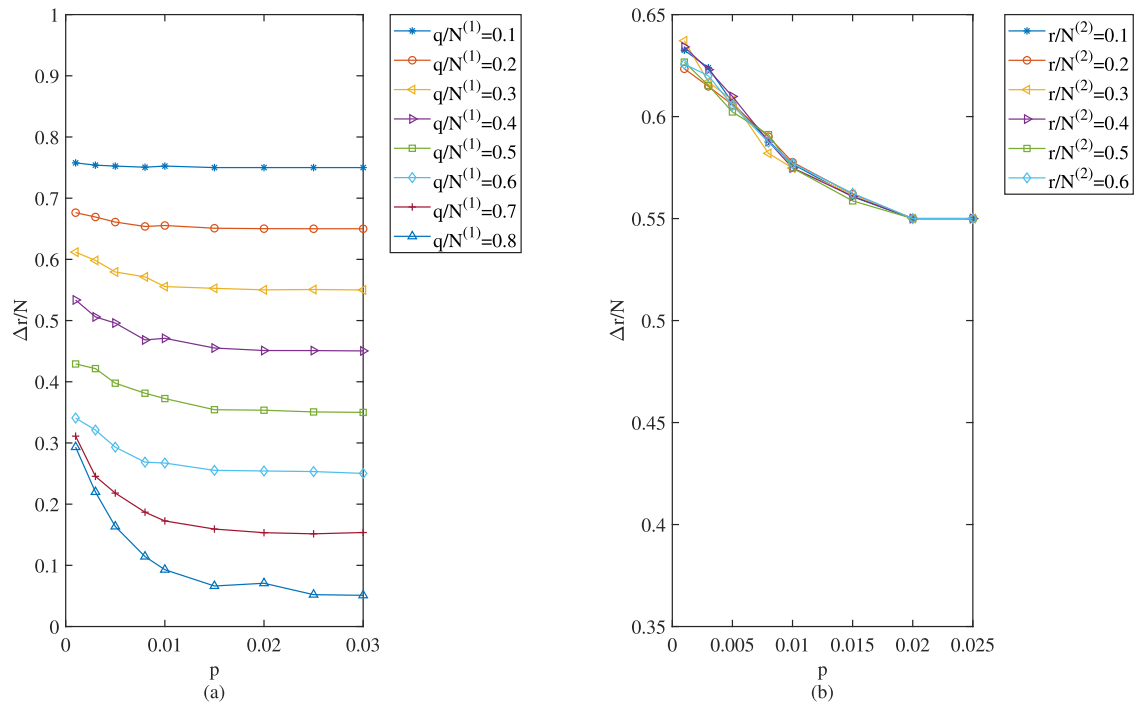


Fig. 6. The minimum number of auxiliary nodes in a two-layer complex dynamical network (SW-SF) (a) The number of auxiliary nodes Δr versus the probability of connection between interlayer nodes with different proportions of the number of sensor nodes. (b) The number of auxiliary nodes Δr versus the probability of connection between interlayer nodes with different proportions of the number of target nodes.

- (3) If the layer where the sensor nodes reside is generated by the SF network, assume that the number of target nodes is fixed, as the probability of interlayer connection p increases, Δr increases to a stable value. When the number of sensors continues to increase to a certain extent, Δr decreases with the increase of p . Assume that the number of sensor nodes is fixed, Δr increases to a peak value and then decrease to a fixed value with the increase of p .
- (4) If the layer where the sensor nodes reside is generated by the SW network, Δr both decreases with the increase of p and finally approaches a fixed value when the number of target nodes or sensor nodes is fixed.

6. Functional state observer design

To reconstruct the target nodes $z(t) = F_0 X(t) \in R^{r_0}$, the following functional state observer structure is proposed in Eq. (7).

$$\begin{cases} \dot{w}(t) = Sw(t) + JY(t) + QU(t), \\ \hat{z}(t) = Dw(t) + EY(t), \end{cases} \quad (7)$$

where $w \in R^\beta$ is the observer state vector, U is the control input, Y is the output vector and $\hat{z}(t) \in R^{r_0}$ is the estimate of $z(t)$. $S \in R^{\beta \times \beta}$, $J \in R^{\beta \times q}$, $Q \in R^{\beta \times m}$, $D \in R^{r_0 \times \beta}$, $E \in R^{r_0 \times q}$ are to be determined such that $\hat{z}(t)$ converges asymptotically to $z(t)$.

The estimate $\hat{z}(t)$ will converges asymptotically to $z(t)$ for any initial condition $w(0)$ if and only if the following condition is satisfied

$$\begin{cases} S \text{ is Hurwitz,} \\ Q = \Gamma\psi, \\ F_0 - D\Gamma - E\xi = 0, \\ S\Gamma + J\xi - \Gamma\varphi = 0. \end{cases} \quad (8)$$

There have been a lot of researches on solving the coefficient matrix S, J, Q, D, E , the observer gain matrix Γ of functional state observer. In this paper, we choose the classical algorithm in Trinh and Fernando [41] and the steps of the algorithm are as follows.

Algorithm 2.

Step 1: Obtain sub-matrices $\varphi_{11}, \varphi_{12}, \varphi_{21}, \varphi_{22}, F_1$ and F_2 by Eq. (9). Check whether the condition in Eq. (10) is satisfied or not. If not, the observer does not exist.

$$P^{-1}\varphi P = \begin{bmatrix} \varphi_{11} & \varphi_{12} \\ \varphi_{21} & \varphi_{22} \end{bmatrix}, \quad P\psi = \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix}, \quad F_0 P = \begin{bmatrix} F_1 & F_2 \end{bmatrix}, \quad (9)$$

where $P = [\xi^+ \ \xi^\perp]$, ξ^+ is the Moore–Penrose inverse of ξ and $\xi^\perp$ is the orthogonal complement of the row space of ξ .

$$\text{rank} \begin{bmatrix} F_2 \varphi_{22} \\ \varphi_{12} \\ F_2 \end{bmatrix} = \text{rank} \begin{bmatrix} \varphi_{12} \\ F_2 \end{bmatrix}. \quad (10)$$

Step 2: Obtain S_1 and S_2 from Eqs. (11), (12). Check the detectability of the pair $(S_2 \ S_1)$. If not, stop as S is not stable.

$$S_1 = (\Phi\Omega^+ \varphi_{12} + F_2\varphi_{22})F_2^+ \quad (11)$$

$$S_2 = (\Omega\Omega^+ - I)\varphi_{12}F_2^+ \quad (12)$$

where $\Omega = \varphi_{12}F_2^\perp$ and $\Phi = -F_2\varphi_{22}F_2^\perp$.

Step 3: Determine an arbitrary matrix $\Theta \in R^{r_0 \times q}$ such that S defined in Eq. (13) is stable.

$$S = S_1 - \Theta S_2 \quad (13)$$

Step 4: Compute Γ_1 from Eq. (14). Obtain $\Gamma_2 = F_2$, $D = I$. Obtain J and E from Eqs. (15), (16) respectively.

$$\Gamma_1 = \Phi\Omega^+ + \Theta(I - \Omega\Omega^+) \quad (14)$$

$$J = \Gamma_1\varphi_{11} + \Gamma_2\varphi_{21} - S\Gamma_1 \quad (15)$$

$$E = F_1 - D\Gamma_1 \quad (16)$$

Step 5: Obtain $Q = \Gamma\psi$, where $\Gamma = [\Gamma_1 \ \Gamma_2]P^{-1}$.

It is easy to check the condition in Eq. (8) is satisfied with all the observer parameters obtained from above steps. This completes the design of a functional observer.

7. Numerical simulations

In this section, some numerical simulations are given to demonstrate the efficiency of the proposed method.

7.1. A two-layer complex dynamical network

Take the two-layer complex dynamical network in Fig. 1 as an example. Its topological structure is shown in Eq. (3). In this model, we consider the dynamics on the first layer is characterized as $A_i^1 = -i$ and on the second layer is characterized as $A_i^2 = i - 7$. If there is a link from node j to node i on the intralayer network, $d_{ij}^l = 1$. If there is a link from node j to node i on the interlayer network, $d_{ij}^{lR} = 2$. The inner coupling $H^l = H^{lR} = 1$. The output matrix ξ can be seen in Section 2 and the input matrix $\psi = [1 \ 2 \ 3 \ 4 \ 5 \ 6 \mid 1 \ 2 \ 3 \ 4 \ 5 \ 6]^T$. The control inputs $U(t) = 1$. In Section 4, node 6 in the first layer have been found as an auxiliary node by using the maximum matching principle, therefore, the augmented node set

$$F_0 = \left[\begin{array}{cccccc|cccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

We design the functional state observer by the algorithm mentioned in Section 6. The estimation performance of the target node with the designed functional observer is shown in Fig. 7. It is obvious that the functional observer has an excellent dynamical behavior and its estimates converge to the target state of the network. The target state estimation error $e(t) = z(t) - \hat{z}(t)$ which can be seen in Fig. 8 also converge to zero.

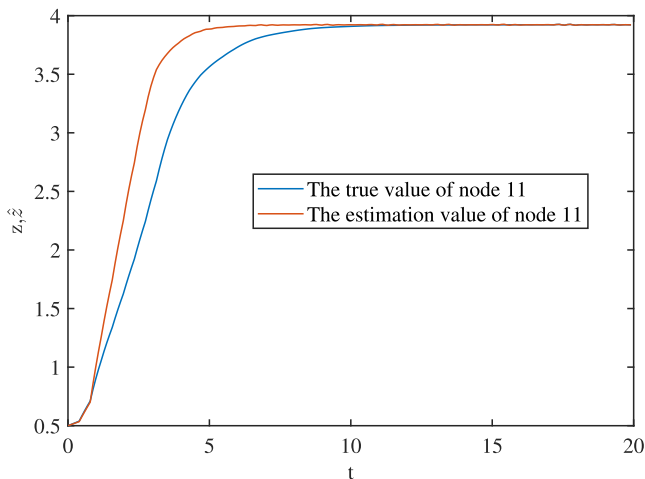


Fig. 7. The estimation performance of the target node in the two-layer complex dynamical network.

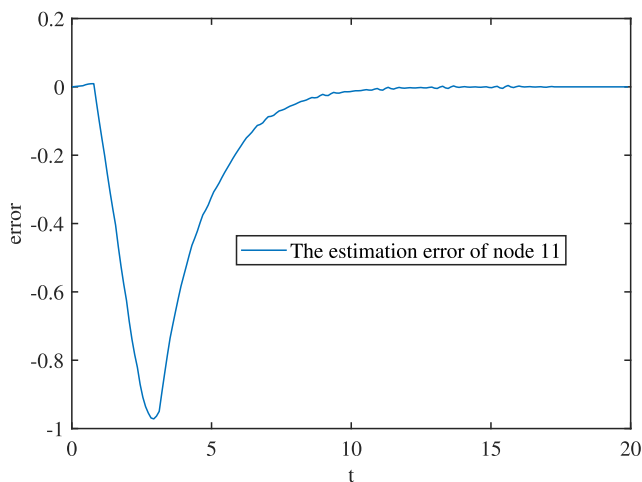


Fig. 8. The estimation error of the target node in the two-layer complex dynamical network.

Through the simulation, we estimate the state information of the target node on the second layer successfully by using the known state information of some nodes on the first layer in a two-layer complex dynamical network. We only need to place observers to the target node and the auxiliary node instead of all unknown nodes, which reduces the number of observers and computational costs.

7.2. A three-layer complex dynamical network

In this subsection, we consider a three-layer complex dynamical network whose structure is shown in Fig. 9. Each layer contains eight nodes. The node dynamics on the first layer is characterized as $A_i^1 = -i$, on the second layer is $A_i^2 = 8 - i$ and on the third layer is

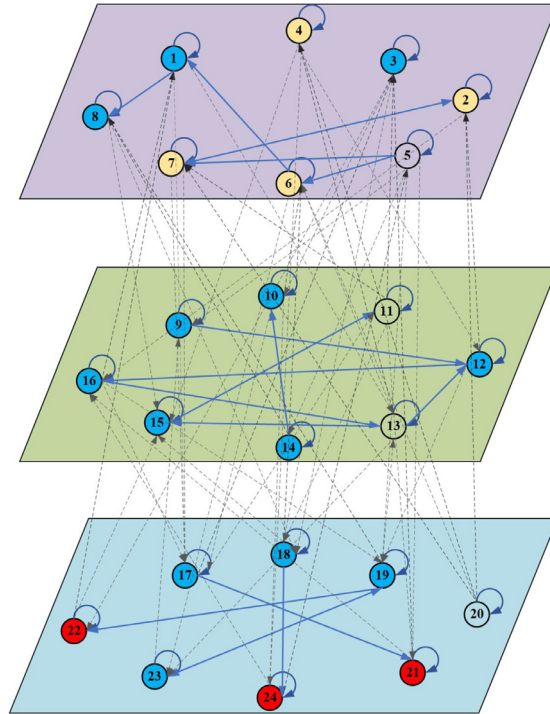


Fig. 9. The topology of the three-layer complex dynamical network. The nodes marked in yellow are the sensor nodes, the nodes marked in red are the target nodes and the nodes marked in blue are the auxiliary nodes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$A_i^3 = 16 - i$. If there is a link from node j to node i on the intralayer network, $d_{ij}^l = 1$. If there is a link from node j to node i on the interlayer network, $d_{ij}^{lR} = 1$. The configuration matrix of the interlayer network is shown in Eqs. (17)–(19). The input matrix $\psi = [1 \ 1 \ \dots \ 1 \ 1]_{1 \times 24}^T$. The control inputs $U(t) = 5$. The node 2, 4, 6 and 7 marked in yellow in the first layer are the sensor nodes. The node 21, 22 and 24 marked in red in the third layer are the target nodes. Our purpose is to realize the state estimation of the target nodes from the proposed method.

$$d_{ij}^{12} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad d_{ij}^{13} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (17)$$

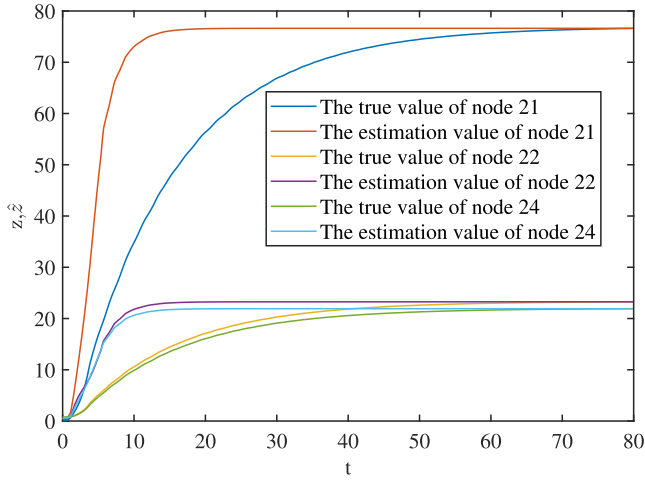


Fig. 10. The estimation performance of the target node in the three-layer complex dynamical network.

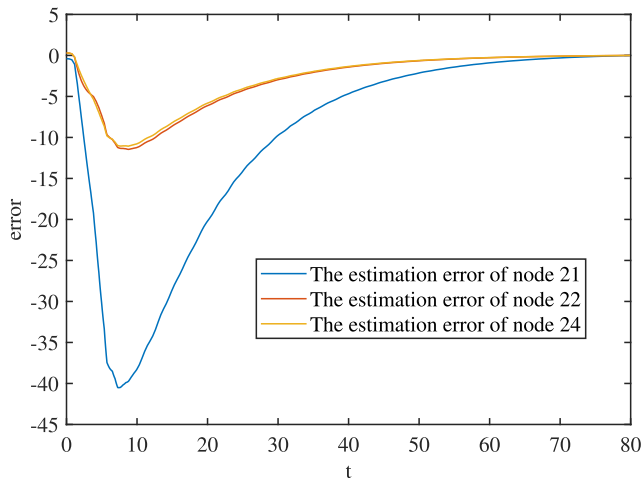


Fig. 11. The estimation errors of the target nodes in the three-layer complex dynamical network.

$$d_{ij}^{21} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad d_{ij}^{23} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}. \quad (18)$$

$$d_{ij}^{31} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad d_{ij}^{32} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (19)$$

The nodes marked in blue in the network are the auxiliary nodes which selected by the maximum matching principle. We only need to place observers to the auxiliary nodes and the target nodes. Fig. 10 shows the estimation performance of the target nodes from the method. As can be seen that the state estimation problem of the target nodes is feasible. The estimation errors are shown in Fig. 11. We can obtain that the estimation errors all converge to zero which means the proposed method is effective.

8. Conclusion

In multilayer complex dynamical networks, it is often impossible to arrange sensors for each node to measure state information. At the same time, many engineering requirements and practical applications only require the estimation of a small set of interested target nodes. In this paper, the problem of state estimation on target nodes in multilayer complex dynamical networks with limited node measurement is studied. Suppose that the state information of some nodes on one layer in a complex multilayer dynamical network can be measured, the auxiliary nodes are selected by using the maximum matching principle of graph theory to construct augmented node set, then a functional state observer is designed by a typical model-based algorithm to realize the state estimation for target nodes in other layers. We also analyze the relationship between interlayer connection probability and the minimum number of the auxiliary nodes in the complex multilayer dynamical network. Compared with full-state observers based on rank condition, the proposed method is based on the idea of graph theory and only requires placing observers to the augmented nodes, which reduces the placement of observers and computation costs. Numerical simulations prove the accuracy of the proposed method. For further research: (1) We will study the optimal sensor placement for minimum-order functional observers. (2) We will extend the relationship between interlayer connection probability and the minimum number of the auxiliary nodes to networks with more than two layers.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Y. Fan, J. Ding, H. Liu, Y. Wang, J. Long, Large-scale multimodal transportation network models and algorithms—Part I: the combined mode split and traffic assignment problem, *Transp. Res. Part E* 164 (2022) 102832.
- [2] Z. Hu, F. Deng, Y. Su, J. Zhang, S. Hu, Security control of networked systems with deception attacks and packet dropouts: a discrete-time approach, *J. Frankl. Inst.* 358 (16) (2021) 8193–8206.
- [3] K.M.A. Kabir, K. Kazuki, T. Jun, The impact of information spreading on epidemic vaccination game dynamics in a heterogeneous complex network—a theoretical approach, *Chaos, Solitons Fractals* 132 (C) (2020) 109548.
- [4] J. Hu, Z. Wang, G.P. Liu, Delay compensation-based state estimation for time-varying complex networks with incomplete observations and dynamical bias, *IEEE Trans. Cybern.* 52 (11) (2022) 12071–12083.
- [5] J. Qian, X. Wang, G.P. Jiang, H. Su, Observer-based semi-global containment of saturated multi-agent systems with uncertainties, *J. Frankl. Inst.* 358 (15) (2021) 7740–7760.
- [6] R. Ding, N. Ujang, H. Hamid, M.S.A. Manan, Y. He, R. Li, J. Wu, Detecting the urban traffic network structure dynamics through the growth and analysis of multi-layer networks, *Physica A* 503 (1) (2018) 800–817.
- [7] J. Hu, C. Jia, H. Liu, X. Yi, Y. Liu, A survey on state estimation of complex dynamical networks, *Int. J. Syst. Sci.* 52 (16) (2021) 3351–3367.
- [8] P.H. Nguyen, W.L. Kling, Distributed state estimation for multi-agent based active distribution networks, in: 2010 IEEE PES General Meeting, 2010, p. 11571351.
- [9] B. Targui, M. Frikel, M. Msaad, S. Safi, Observer design for the state estimation of a class of communication networks, in: 18th Mediterranean Conference on Control and Automation, 2010, p. 11474011.
- [10] C.X. Fan, Y.H. Wan, G.P. Jiang, Topology identification for a class of complex dynamical networks using output variables, *Chin. Phys. B* 21 (2) (2012) 020510.
- [11] X. Wu, G.P. Jiang, X. Wang, State estimation for general complex dynamical networks with packet loss, *IEEE Trans. Circuits Syst. II* 65 (11) (2017) 1753–1757.
- [12] X. Wang, H. Su, F. Zhang, C. Chen, A robust distributed interval observer for LTI systems, *IEEE Trans. Autom. Control* 68 (3) (2023) 1337–1352.
- [13] X. Wang, G. Jiang, X. Wu, A novel method of topology identification for general complex dynamic networks with incomplete measurements, *Int. J. Mod. Phys. C* 29 (5) (2018) 1–14.
- [14] X. Wu, X. Wang, M. Gu, G.P. Jiang, Structural controllability for a class of complex networks with root strongly connected components, *IEEE Access* 6 (2020) 123986–123995.
- [15] X. Wan, Y. Li, Y. Li, M. Wu, Finite-time H_∞ state estimation for two-time-scale complex networks under stochastic communication protocol, *IEEE Trans. Neural Netw. Learn. Syst.* 33 (1) (2022) 21525174.
- [16] X. Wang, X. Wang, H. Su, J. Lam, Reduced-order interval observer based consensus for MASs with time-varying interval uncertainties, *Automatica* 135 (2022) 109989.
- [17] W. Yu, G. Chen, J. Lv, On pinning synchronization of complex dynamical networks, *Automatica* 45 (2) (2009) 429–435.
- [18] C.X. Fan, G.P. Jiang, F.H. Jiang, Synchronization between two complex dynamical networks using scalar signals under pinning control, *IEEE Trans. Circuits Syst. I* 57 (11) (2010) 2991–2998.
- [19] A.E. Motter, Networkcontrollogy, *Chaos* 25 (9) (2015) 097621.
- [20] N. Xue, A. Chakraborty, Control inversion: a clustering-based method for distributed wide-area control of power systems, *IEEE Trans. Control Netw. Syst.* 6 (3) (2019) 937–949.
- [21] Y. Liu, Z. Wang, Y. Yuan, F.E. Alsaadi, Partial-nodes-based state estimation for complex networks with unbounded distributed delays, *IEEE Trans. Neural Netw. Learn. Syst.* 29 (8) (2017) 3906–3912.
- [22] J. Li, H. Dong, Z. Wang, X. Bu, Partial-neurons-based passivity-guaranteed state estimation for neural networks with randomly occurring time delays, *IEEE Trans. Neural Netw. Learn. Syst.* 31 (9) (2019) 3747–3753.
- [23] Y.Y. Liu, J.J. Slotine, A. L. Barabási, Controllability of complex networks, *Nature* 473 (7346) (2011) 167–173.
- [24] A.N. Montanari, C. Duan, L.A. Aguirre, A.E. Motter, Functional observability and target state estimation in large-scale networks, *Proc. Natl. Acad. Sci.* 119 (1) (2022) e2113750119.
- [25] R. Berner, V. Mehrmann, E. Schöll, S. Yanchuk, The multiplex decomposition: an analytic framework for multilayer dynamical networks, *SIAM J. Appl. Dyn. Syst.* 20 (4) (2021) 901–906.
- [26] S. Funk, E. Gilad, C. Watkins, The spread of awareness and its impact on epidemic outbreaks, *Proc. Natl. Acad. Sci.* 106 (16) (2009) 6872–6877.
- [27] J. Aguirre, R.S. Escobaza, R. Gutierrez, D. Papo, J.M. Buldu, Synchronization of interconnected networks: the role of connector nodes, *Phys. Rev. Lett.* 112 (24) (2014) 248701.

- [28] R. Li, W. Yu, J. Lv, Synchronization on complex networks of networks, *IEEE Trans. Neural Netw. Learn. Syst.* 25 (11) (2014) 2110–2118.
- [29] S. Gómez, A.D. Guílera, J.G. Gardeñes, C.J.P. Vicente, Y. Moreno, A. Arenas, Diffusion dynamics on multiplex networks, *Phys. Rev. Lett.* 110 (2) (2013) 028701.
- [30] S. Boccaletti, G. Bianconi, R. Criado, C.I. Genio, J.G. Gardeñes, M. Romance, I.S. Nadal, Z. Wang, M. Zanin, The structure and dynamics of multilayer networks, *Phys. Rep.* 544 (1) (2014) 1–122.
- [31] G. Mei, X. Wu, Y. Wang, M. Hu, J. Lu, G. Chen, Compressive-sensing-based structure identification for multilayer networks, *IEEE Trans. Cybern.* 48 (2) (2018) 754–764.
- [32] A. Chapman, M.N. Abdolyousefi, M. Mesbahi, Controllability and observability of network-of-networks via cartesian products, *IEEE Trans. Autom. Control* 59 (10) (2014) 2668–2679.
- [33] Y.Q. Hao, Q.Y. Wang, Z.S. Duan, G.R. Chen, Controllability of Kronecker product network, *Automatica* 110 (2019) 108597.
- [34] C. Chen, A. Surana, A. Bloch, Controllability of hypergraphs, *IEEE Trans. Netw. Sci. Eng.* 8 (2) (2021) 1646–1657.
- [35] J.N. Wu, X. Li, G.R. Chen, Controllability of deep-coupling dynamical networks, *IEEE Trans. Circuits Syst. I* 67 (12) (2020) 5211–5222.
- [36] L.X. Jiang, T.K. Tang, J.H. Lv, Controllability of multilayer networks, *IEEE Trans. Circuits Syst. I* 10 (2021) 1–11.
- [37] L.Q. Cao, L.F. Wang, Z. Kong, Y. Guo, Controllability of multi-layer heterogeneous complex network systems, *Acta Autom. Sin.* 48 (x) (2021) 1–13.
- [38] T.L. Fernando, H.M. Trinh, L. Jennings, Functional observability and the design of minimum order linear functional observers, *IEEE Trans. Autom. Control* 55 (5) (2010) 1268–1273.
- [39] J. Gao, Y.Y. Liu, R.M.D. Souza, A.L. Barabási, Target control of complex networks, *Nat. Commun.* 5 (2014) 5415.
- [40] F. Rotella, I. Zambettakis, A note on functional observability, *IEEE Trans. Autom. Control* 61 (10) (2016) 3197–3202.
- [41] H. Trinh, T. Fernando, *Functional Observers for Dynamical Systems*, Springer, Berlin, 2012.
- [42] M. Darouach, Existence and design of functional observers for linear systems, *IEEE Trans. Autom. Control* 45 (5) (2000) 940–943.
- [43] H. Trinh, S. Nahavand, T. Tran, Algorithms for designing reduced-order functional observers of linear systems, *Int. J. Innov. Comput., Inf. Control* 4 (2) (2008) 321–333.
- [44] C.T. Lin, Structural controllability, *IEEE Trans. Autom. Control* 19 (3) (1974) 201–208.